

Exercise 1: Word Frequency Counter

Objective

Count how many times each word appears in a sentence.

Input

```
text = "I like NLP I like Python NLP is fun"
```

Expected Output

```
I : 2
```

```
like : 2
```

```
NLP : 2
```

```
Python : 1
```

```
is : 1
```

```
fun : 1
```

Tasks

- Split the sentence into words.
- Count each word.
- Store the result in a dictionary.
- Print the dictionary.

Concepts Learned

- Strings
 - Lists
 - Dictionary
 - Term Frequency (TF)
-

Exercise 2: Vocabulary Builder

Objective

Find all unique words in a corpus.

Input

```
corpus = [  
    "I like NLP",  
    "I like AI",  
    "AI is amazing"  
]
```

Expected Output

Vocabulary

AI

I

NLP

amazing
is
like

Vocabulary Size = 6

Tasks

- Read every sentence.
- Split into words.
- Create a vocabulary using a set.
- Print the vocabulary size.

Concepts Learned

- Vocabulary
- Unique words

Exercise 3: Build a Word Co-occurrence Matrix

Objective

Construct a simple co-occurrence matrix using a window size of 1.

Input

sentence = "the cat sat on the mat"

Example Output

Word the cat sat on mat

the	0	1	0	1	1
cat	1	0	1	0	0
sat	0	1	0	1	0
on	1	0	1	0	1
mat	1	0	0	1	0

Tasks

- Build vocabulary.
- Create an empty matrix.
- Count neighboring words.

Concepts Learned

- Co-occurrence Matrix
- Context Window

Exercise 4: Detect Data Sparsity

Objective

Find word pairs that never occur together.

Input

Use the co-occurrence matrix from Exercise 3.

Expected Output

(cat, mat) -> Never occurred

(sat, the) -> Never occurred

(cat, on) -> Never occurred

Tasks

- Traverse the matrix.
- Find all cells with value 0.
- Print those word pairs.

Concepts Learned

- Data Sparsity
- Sparse Matrix

SPARSE MATRIX

Exercise 5: Implement Laplace Smoothing

Objective

Calculate word probabilities using Laplace smoothing.

Input

```
word_counts = {  
    "cat":4,  
    "dog":3,  
    "bird":1  
}
```

Formula

$$P(w) = \frac{\text{Count} + 1}{\text{Total} + \text{Vocabulary}}$$

Expected Output

cat : 0.50

dog : 0.40

bird : 0.20

(Values will depend on the exact counts and denominator.)

Tasks

- Compute total words.

- Compute vocabulary size.
- Apply Laplace smoothing.
- Print probabilities.

Concepts Learned

- Probability
- Laplace Smoothing

Exercise 6: Simple Keyboard Prediction

Objective

Predict the most common next word.

Training Corpus

I like NLP

I like Python

I like coffee

I love NLP

Input

I like

Expected Output

Suggestions

NLP

Python

coffee

Tasks

- Build bigram counts.
- Store next-word frequencies.
- Display the top three predictions.

Concepts Learned

- Bigram Language Model
- Keyboard Prediction

Top N Probabilities prediction : Top 3 Predict

$$P(\text{NLP/like}) = 0.4$$

$$P(\text{PYthon/like}) = 0.3$$

$$P(\text{Coffe/like}) = 0.2$$

0.01 0.02 0.03 0.04 0.05

Exercise 7: Calculate TF-IDF

Objective

Compute TF, IDF, and TF-IDF scores.

Corpus

Document 1:

I like NLP

Document 2:

I like AI

Document 3:

AI is amazing

Tasks

- Calculate TF.
- Calculate DF.
- Calculate IDF.
- Compute TF-IDF.

DF: In how many documents that specific word is found?

Concepts Learned

- TF
- DF
- IDF
- TF-IDF

Exercise 8: Dense Vector Similarity

Objective

Compute cosine similarity between two dense vectors.

Input

cat = [0.4, 0.2, 0.8]

dog = [0.5, 0.1, 0.7]

Expected Output

Cosine Similarity = 0.98

(Approximate value.)

Tasks

- Compute dot product.
- Compute vector magnitudes.
- Calculate cosine similarity.

Concepts Learned

- Dense Vectors
- Cosine Similarity

Exercise 9: Find Similar Words

Objective

Find the word most similar to a given word.

Input

embeddings = {

"cat":[0.4,0.2,0.8],

"dog":[0.5,0.1,0.7],

"car":[-0.7,0.8,-0.2],

"lion":[0.45,0.25,0.82]

}

Expected Output

Most similar to cat

lion

Similarity = 0.99

(Approximate value.)

Tasks

- Compare the target vector with all others.
- Compute cosine similarity.
- Print the most similar word.

Concepts Learned

- Word Embeddings
- Semantic Similarity

Exercise 10 (Assignment) : Mini NLP Pipeline

Objective

Combine multiple concepts into one application.

Input

I love NLP

I love AI

AI loves Python

Python loves data

Tasks

Your program should:

1. Build the vocabulary.
2. Count word frequencies.
3. Compute TF.
4. Build the co-occurrence matrix.
5. Detect sparse entries.
6. Apply Laplace smoothing to unigram probabilities.
7. Predict the next word after a user-entered word.
8. Display the top three most frequent words.

Example Output

Vocabulary Size : 7

Top Words

love : 2

AI : 2

Python : 2

Prediction after "love"

NLP

AI

Sparse Pairs

(NLP,data)

(I,Python)

Exercise 11: Predict the Next Word Using Conditional Probability

Objective

Use conditional probability to predict the most likely next word after a given word using a small corpus.

Background

The conditional probability of a word w_2 given a previous word w_1 is:

$$P(w_2 | w_1) = \frac{\text{Count}(w_1, w_2)}{\text{Count}(w_1)}$$

where:

- $\text{Count}(w_1, w_2)$ = Number of times the bigram (w_1, w_2) appears.
 - $\text{Count}(w_1)$ = Number of times w_1 appears as the first word in a bigram.
-

Training Corpus

I like NLP

I like Python

I like coffee

I love NLP

You like coffee

Step 1: Bigram Counts

Bigram	Count
I → like	3
I → love	1
You → like	1
like → NLP	1
like → Python	1
like → coffee	2
love → NLP	1

Step 2: Compute Conditional Probabilities

Example 1

Compute:

$$P(\text{coffee} | \text{like})$$

The word like is followed by:

- NLP (1 time)
- Python (1 time)
- coffee (2 times)

Total occurrences after like:

$$1 + 1 + 2 = 4$$

Therefore,

$$P(\text{coffee} \mid \text{like}) = \frac{2}{4} = 0.50$$

Example 2

$$P(\text{Python} \mid \text{like}) = \frac{1}{4} = 0.25$$

Example 3

$$P(\text{NLP} \mid \text{like}) = \frac{1}{4} = 0.25$$

Programming Task

Write a Python program that:

1. Reads the training corpus.
2. Builds bigram counts.
3. Calculates conditional probabilities for every bigram.
4. Asks the user to enter a word.
5. Displays all possible next words and their conditional probabilities.
6. Predicts the word with the highest conditional probability.

Sample Input

Enter a word: like

Expected Output

Possible next words:

coffee : 0.50

NLP : 0.25

Python : 0.25

Predicted next word: coffee

Bonus Challenge

Modify your program so that it displays the top three predictions instead of only the best one.

Example:

Top Predictions

1. coffee (0.50)
2. NLP (0.25)
3. Python (0.25)